Answers and talk notes

Simplify Algebraic Expressions

What your student is learning

Tonight we simplify an expression by combining like terms, then evaluate the simplified version for a given number.

Answers

Ask your student to explain before you look.

1. **A.** 2n. 2n has the same variable as 6n, so 6n + 2n = 8n.
2. **B.** 5w. w means 1w, so 4w + 1w = 5w.
3. **B.** 8c + 7. 6c + 2c = 8c and 3 + 4 = 7, so 8c + 7.
4. **D.** 23. 5 × 4 = 20, and 20 + 3 = 23.
5. **B.** 14. Simplify to 4z + 6. When z = 2, 4 × 2 + 6 = 14.
6. **C.** 7x + 3. 7x + 3 has one x term and one plain number, so nothing more can combine.

Talk about it

Which terms are alike? Do the long and the short expression give the same number when x is replaced?

Home mission

Pick x = 4 for the price of one shirt. Ask your student to find 5x + 7 + 3x + 2 two ways: simplify first, or put in 4 first. Do the answers match?

When they are stuck, ask

- Which terms have the same letter part?
- Does this number have a variable attached to it?
- Does the original expression give the same value as the simplified one?

Try not to say

- Add all the numbers together, that is simplifying.
- 3x + 2x is 5x squared.
- The plain number can join the x-term if it is close by.